System Validation Report

for

Student Attendance Management System (SAMS)

Version: 1.0
Date: 06-11-2026

Team 1

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1. Introduction

This report gives the results of system testing of SAMS against the test cases TC-01 to TC-17 and the security checks in section 5.1 of the Software Test Plan. Each test case was run end to end through the web interface in a browser, the same way a user would use it, and the result was checked on screen and in the database.

2. Test Environment

Item	Details
Build	main at commit c1edd6d (all Sprint 1 and Sprint 2 stories merged)
Server	Django development server, SQLite database, console email backend (mails printed to the server log)
Browser	Chromium (headless), 1280 x 800, driven by a Playwright script that clicks through each test case and saves a screenshot
Test data	1 admin, 1 faculty (Ravi Kumar), class CSE 5 C with the 4 team members, class CSE 5 D with 70 students, subject UE24CS341A, 3 days of earlier attendance
Date run	02-11-2026

3. Summary

Test cases	Passed	Failed	Not run
17	17	0	0

Security checks (Test Plan 5.1)	Passed
9	9

4. Test Case Results

TC	Req	Test	Expected	Actual result	Result
TC-01	REQ-1	Login with valid ID and password	Dashboard opens	Redirected to /accounts/dashboard/ with faculty dashboard	Pass
TC-02	REQ-1	Login with wrong password	"Invalid username or password" shown	Stayed on login page, message 'Invalid username or password.' shown	Pass
TC-03	REQ-2, NFR-7	Student tries to open admin page	403 Forbidden	Student opening admin upload page got HTTP 403	Pass
TC-04	REQ-4	Upload Excel with 10 students (2	8 added, 2 rejected	8 student(s) added, 2 row(s) rejected;	Pass

		without SRN)	with reason	rejected rows listed with reason	
TC-05	REQ-5, REQ-6	Admin adds subject and class	Saved and listed	Admin added subject UE24CS342A through /admin, it is listed	Pass
TC-06	REQ-8	Mark attendance with 3 absentees	Records saved, confirmation	Class of 70 students: 67 Present, 3 Absent saved, confirmation shown	Pass
TC-07	REQ-9	Mark same subject, date and hour again	"Attendance already marked"	Warning 'Attendance already marked for this subject, date and hour.' shown, no new rows	Pass
TC-08	REQ-10	Edit attendance within 3 days and after 3 days	Within: saved and logged; after: blocked	Same-day edit saved and logged (1 log row); record marked 4 days ago blocked for faculty	Pass
TC-09	REQ-11	% after 3 of 4 classes attended	75.0% shown	Santosh attended 3 of 4 classes (one edited to Absent in TC-08): 75.0% shown	Pass
TC-10	REQ-12, NFR-8	Student views own attendance, tries other pages	Own data only; others 403	/reports/my/ shows only own subjects (no student id in URL); class report HTTP 403, shortage list HTTP 403 for student	Pass
TC-11	REQ-13, REQ-14	Report for date range, PDF and Excel	Report shown, both files download	Report shown for date range; downloaded attendance_UE24CS341A_20261003_20261102.pdf (2 KB), attendance_UE24CS341A_20261003_20261102.xlsx (5 KB)	Pass
TC-12	REQ-15	Shortage list	Only students below 75% listed	Subhransi (below 75%) listed; Gyaan (100%) not listed	Pass
TC-13	REQ-16	Attendance drops below 75%	Shortage email sent	4 'Attendance shortage' mails printed by the console email backend right after attendance was saved, to: pes1ug24cs161@sams.local, s0@sams.local, s1@sams.local, s2@sams.local. No mail to students at or above 75%	Pass
TC-14	REQ-17 to 19	Apply leave, faculty approves	Approved; days marked Leave; % updated	Request Pending -> Approved; 2 absent days set to Leave; Subhransi % 50.0 -> 100.0	Pass
TC-15	NFR-1, NFR-2, NFR-3	Timing of page load, save and report	< 3 s, < 2 s, < 5 s	Load class list (70): 0.13s; save 70 students: 0.25s; dashboard: 0.05s; semester report: 0.07s (single user, local)	Pass
TC-16	NFR-6	Password stored in database	Hashed, not plain text	accounts_user.password = 'pbkdf2_sha256$600000$Dp3jnum...' (hashed, not plain text)	Pass
TC-17	NFR-9	Idle for 31 minutes	Session expired, back to login	SESSION_COOKIE_AGE = 1800 s (30 min), saved on every request. Session expiry moved into the past to simulate 31 min idle; next request	Pass

				redirected to login	

5. Security Validation (Test Plan 5.1)

Check	How it was tested	Result
Password hashing (NFR-6)	Read accounts_user.password from the database (TC-16)	Pass
Role based access (NFR-7)	Student on admin page, faculty on student page, student on faculty reports	Pass
Student data privacy (NFR-8)	Student pages use the logged in user only; other report pages return 403 (TC-10)	Pass
Session timeout (NFR-9)	Session expired after idle time, redirected to login (TC-17)	Pass
CSRF protection (NFR-10)	Login form submitted with a wrong CSRF token	Pass
SQL injection (NFR-10)	Entered ' OR 1=1 -- as username and password	Pass
Admin site access	Faculty opening /admin/ is sent to the admin login (not staff)	Pass
Audit log (NFR-11)	Edit of a record created an AttendanceLog row with user and time (TC-08)	Pass
Sensitive data in logs (SO-3)	Searched the server log for the test passwords - none found	Pass

6. Performance (NFR-1 to NFR-3)

Measurement	Result	Limit	Result
Single user timings (TC-15)	Load class list (70): 0.13s; save 70 students: 0.25s; dashboard: 0.05s; semester report: 0.07s (single user, local)	3 s / 2 s / 5 s	Pass
100 users at the same time (NFR-1)	100 concurrent logged-in requests to /reports/my/: 100 OK, median 0.67s, 95th 1.08s, max 1.21s	3 s per page	Pass

Note: measured on a laptop with the Django development server and SQLite, not a production server. Numbers are well inside the limits but should be read as indicative.

7. Defects

No defects were found during system testing. Defects found earlier (D-1 to D-4, see Unit Test Report section 5) were fixed in PR #36 and retested here (TC-03, TC-05, TC-12, TC-13 pass).

Appendix A: Screenshots

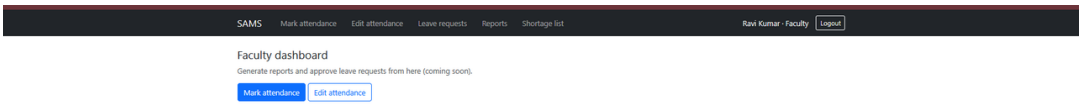


Figure: TC-01 Faculty dashboard after login

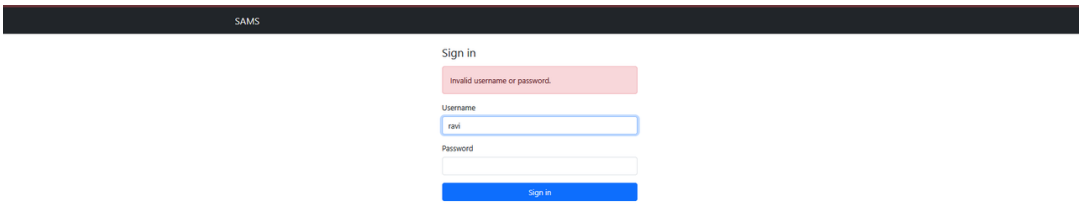


Figure: TC-02 Wrong password

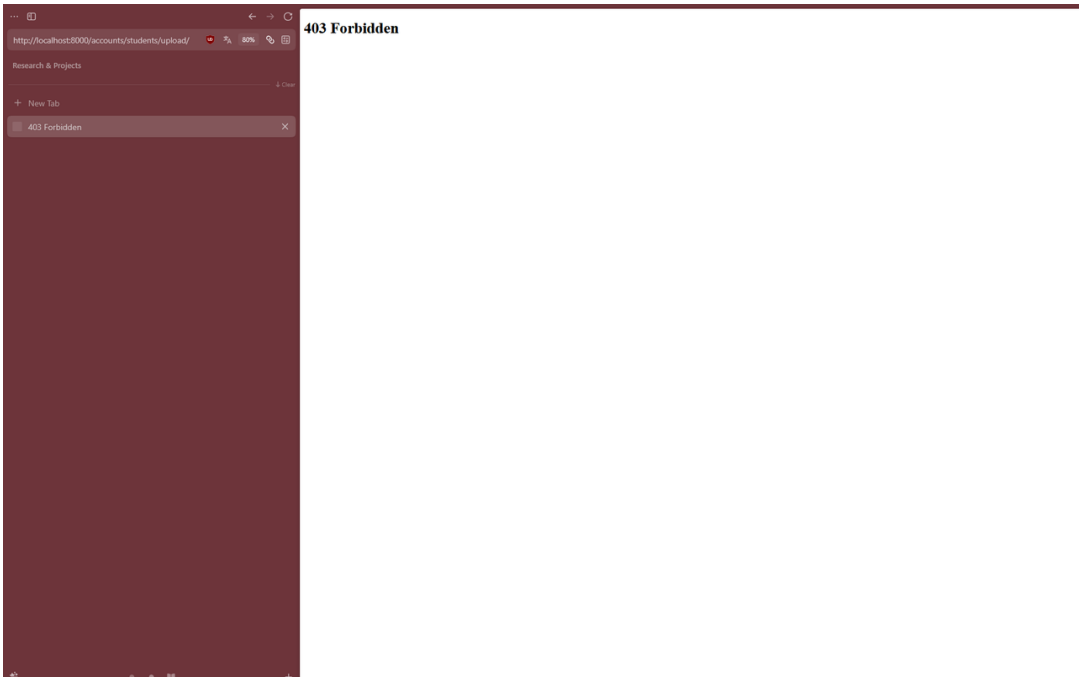


Figure: TC-03 Student blocked from admin page

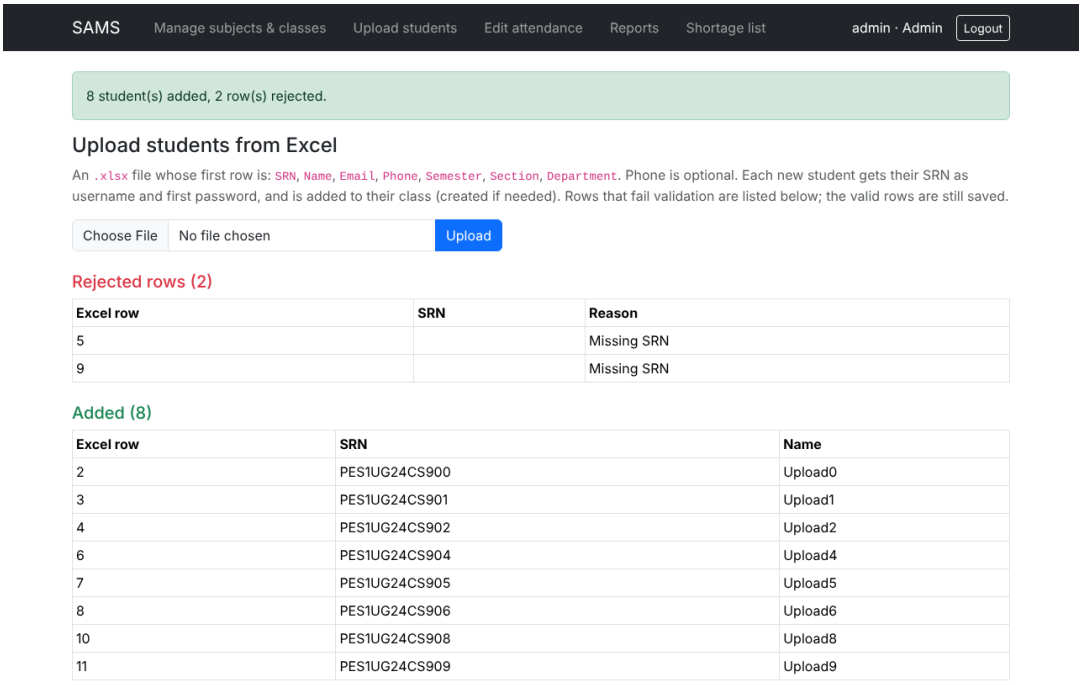


Figure: TC-04 Excel upload result

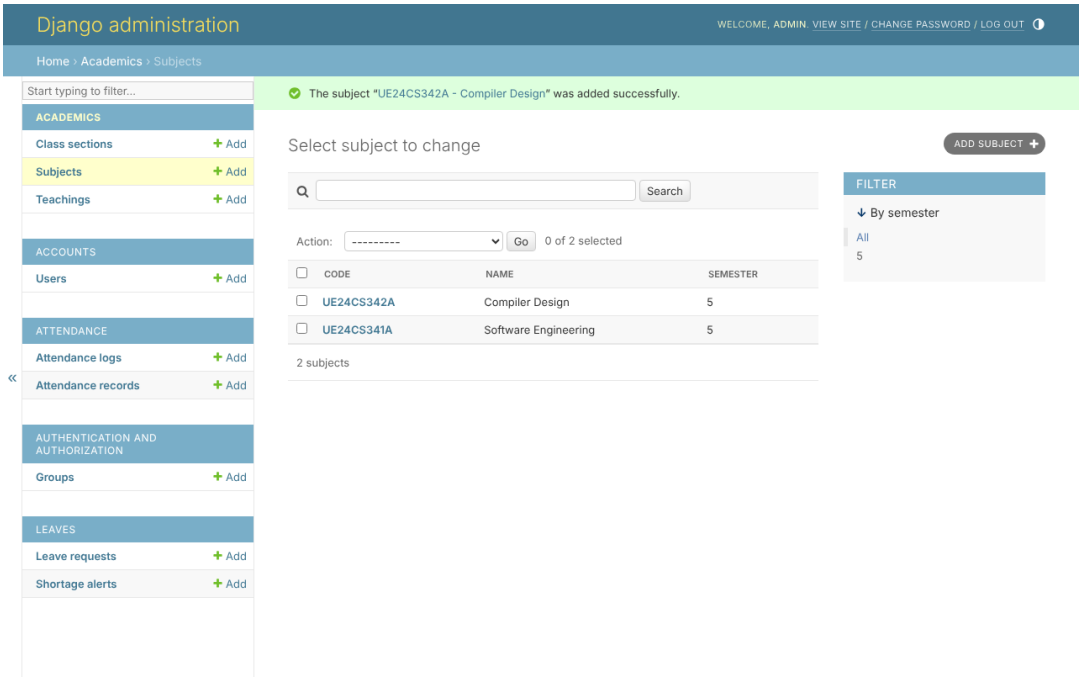


Figure: TC-05 Subject added in admin

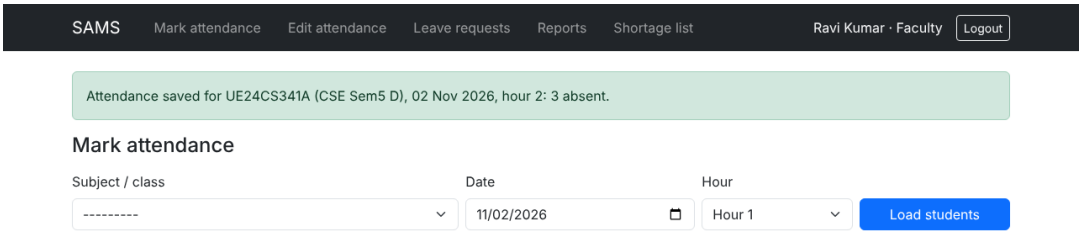


Figure: TC-06 Attendance saved for 70 students

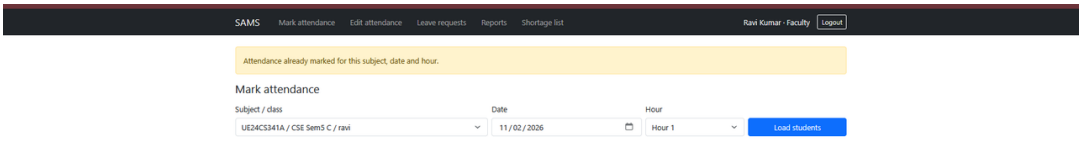


Figure: TC-07 Duplicate marking blocked

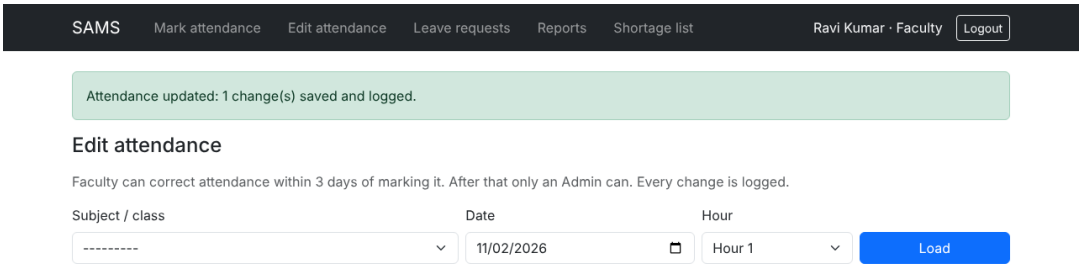


Figure: TC-08 Edit within 3 days saved

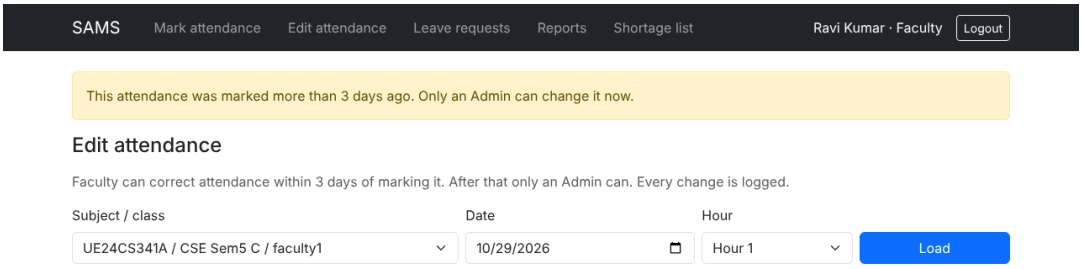


Figure: TC-08 Edit after 3 days blocked

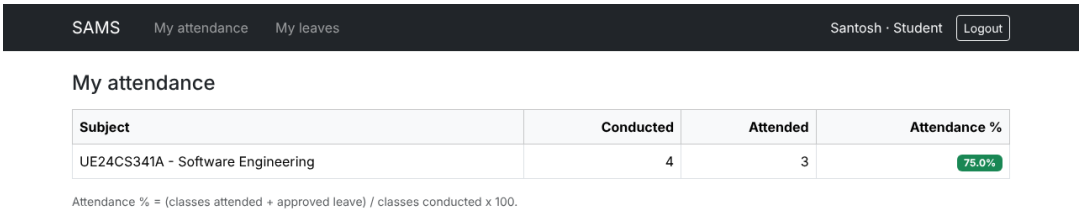


Figure: TC-09 Student percentage 75.0%

403 Forbidden

Figure: TC-10 Student blocked from faculty report

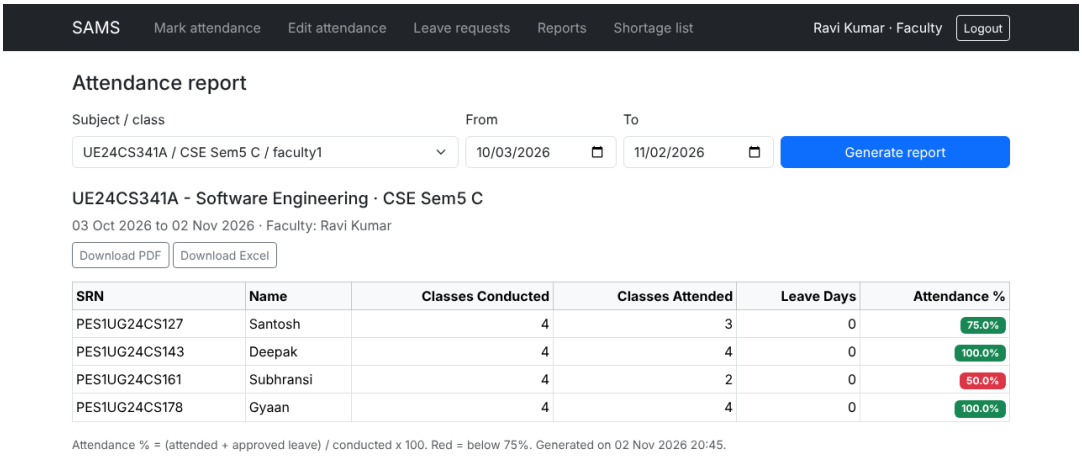


Figure: TC-11 Class report with export buttons

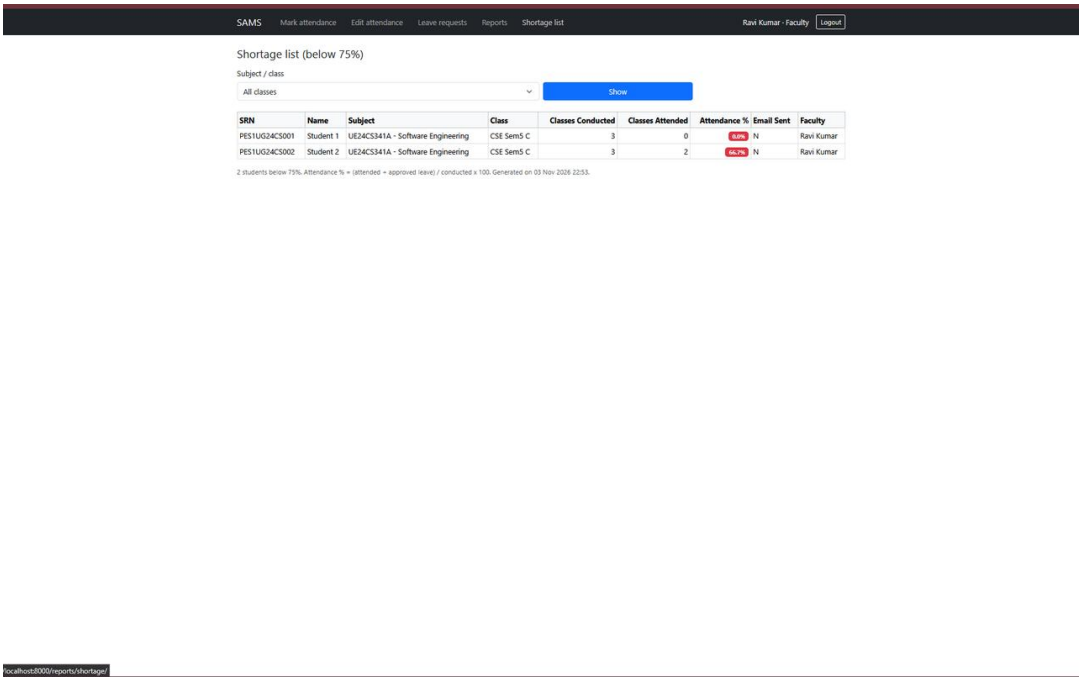


Figure: TC-12 Shortage list

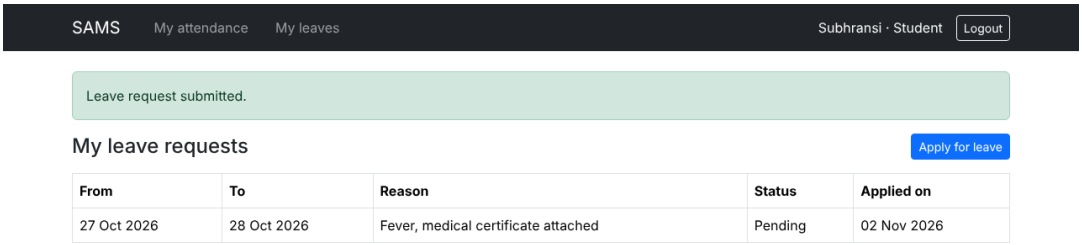


Figure: TC-14 Leave applied

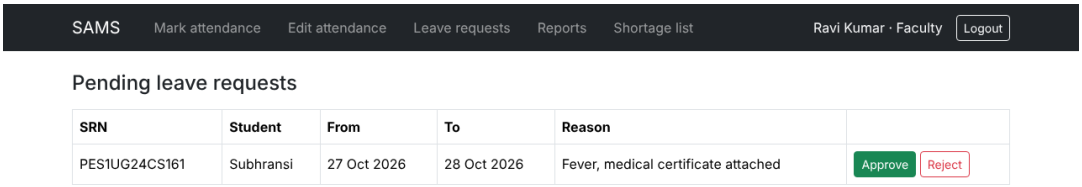


Figure: TC-14 Pending request (faculty)

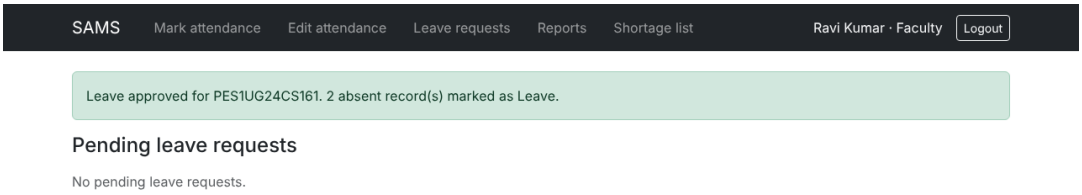


Figure: TC-14 Leave approved

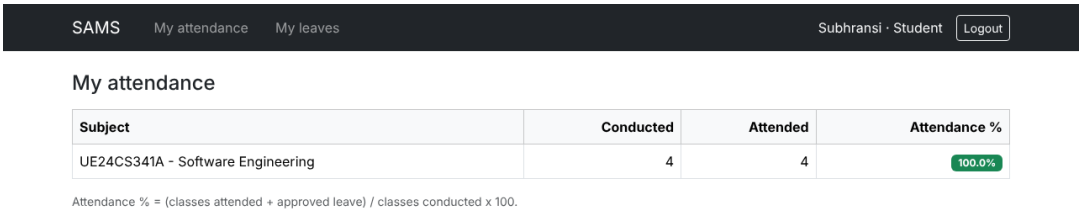


Figure: TC-14 Student percentage after approval

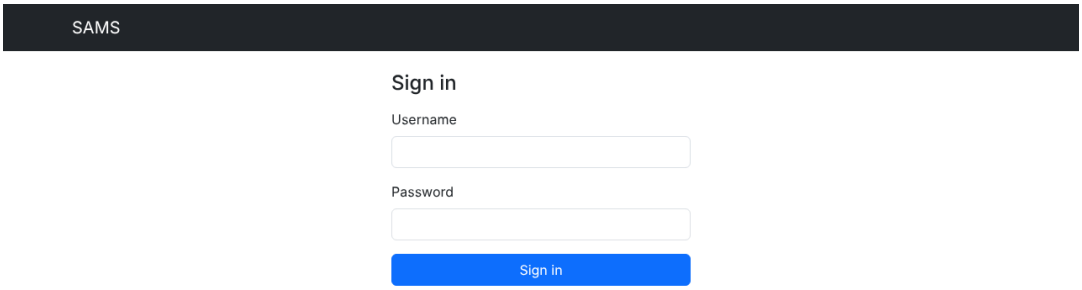


Figure: TC-17 Session expired, login page

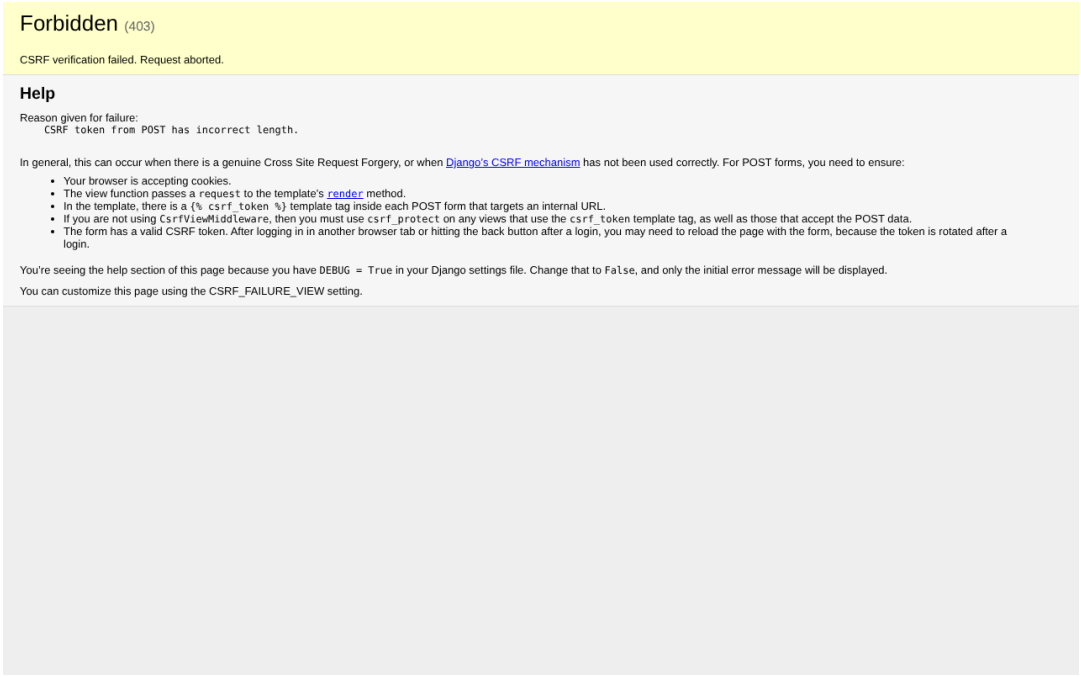


Figure: Security: wrong CSRF token rejected